Assignment 2: Effect of Alloying and Temperature on Fault Energies

Course: MM309N – Computational Methods of Materials
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Group 2 – Alloy System: Al–Co–Fe

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1. Introduction / Background

Stacking fault energy **(SFE)** is a very important material property that determines the behavior of **dislocations, the deformation process, and the mechanical behavior** of crystalline materials. SFE determines the dominance of **partial dislocations, cross-slip, and twinning** in face-centered cubic (FCC) system, and it is attributed to ductility and work-hardening behavior. The process of alloying and temperature dramatically changes SFEs with the changes in electronic structure and lattice distortions.
This Assignment examined the ternary alloy system of **Al-Co-Fe** at three temperatures, i.e.: **400 K, 200 K,** and **650 K,** to determine how composition and temperature influence (γ_{ISF}), extrinsic (γ_{ESF}), and twin (γ_{Twin}) fault energies. SFE was calculated with the **Diffuse Multi-Layer Fault (DMLF) model,** as presented in Charpagne et al. (2020) [1]., and compared with the **Axial Next-Nearest Neighbor Ising (ANNNI) model** to understand the physical implications of both approaches.

2. Simulation Methodology and Details

- The **LAMMPS** molecular dynamics package was used to perform atomistic simulations.
- The **NIST Interatomic Potentials Repository** provided potentials of Embedded Atom Method (EAM) of **Al, Co,** and **Fe.**
- The potentials were **benchmarked** through a comparison between the equilibrium **lattice constants** of pure elements and literature results.
- **FCC, HCP,** and **DHCP** configurations were built as supercells, so that the box dimensions were similar to prevent directional bias.
- Under **periodic boundary conditions**, simulations were conducted the simulations in all three directions with `metal units` and the `atom atomic style`.
- The equilibrating systems were the **NPT** ensemble at the desired temperatures.
- The `set type/fraction` was used to place the **Al, Co,** and **Fe** atoms at random to reflect the ternary alloy composition.
- The stacking fault energies were obtained using the **DMLF model** equations:

$$\gamma_{\mathrm{ISF}} = \frac{4(E_{\mathrm{DHCP}} - E_{\mathrm{FCC}})}{A_{\mathrm{FCC}}}$$

$$\gamma_{\mathrm{ESF}} = \frac{E_{\mathrm{HCP}} + 2E_{\mathrm{DHCP}} - 3E_{\mathrm{FCC}}}{A_{\mathrm{FCC}}}$$

$$\gamma_{\mathrm{Twin}} = \frac{2(E_{\mathrm{DHCP}} - E_{\mathrm{FCC}})}{A_{\mathrm{FCC}}}$$

- **OVITO** was used to do visualization and post-processing, and ternary compositional maps were done in Python with **mpltern** (Figure-2).

3. Results and Discussion

3.1 Benchmarking Results

Benchmarking confirmed close agreement with literature values, with lattice parameters for Al, Co, and Fe deviating by **0.6% to 12% from reference data**, validating the chosen potential set (Figure-1).

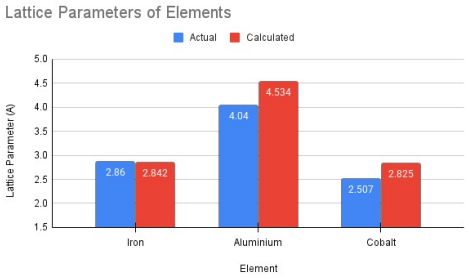
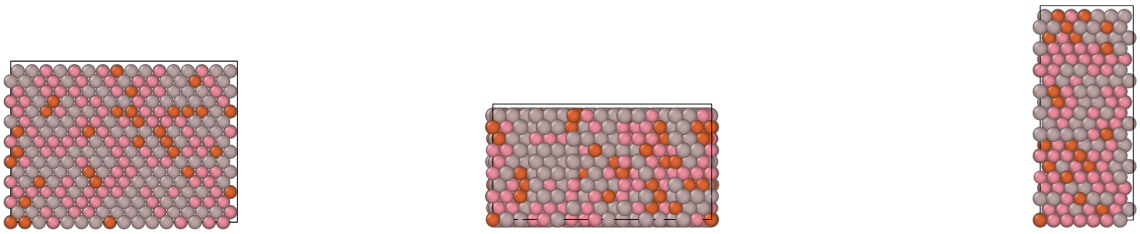


Figure 1: Benchmarking of Al, Co, Fe



(a) FCC structure (b) HCP structure (c) DHCP structure

Figure 2: Visualization of FCC, HCP, and DHCP structures for the Al–Co–Fe alloy system at composition 12

3.2 Cohesive Energy Variation

Addition of Fe causes cohesive energy to become **more negative** meaning stronger bonding strength. Al additions **increase** the cohesive energy (reducing it to less negative) signifying a weaker interatomic attraction.

3.3 Stacking Fault Energies (DMLF Model)

As we can see in Figure-3;

- Both (γ_{ISF}) and (γ_{Twin}) are **temperature-dependent**, indicating that an increased temperature promotes the stabilization of the perfect FCC stacking and the decreased tendency of twinning.
- There is **composition-sensitive variation** in (γ_{ESF}), which is lowest around Co-rich compositions, suggesting the facility of **easy crack formation** of twin faults.
- The Al-rich alloys have **low SFEs,** which is in line with the expansion in the lattice and low cohesive strength.

3.4 Comparison Between DMLF and ANNNI Models

As we can see in Figure-4;

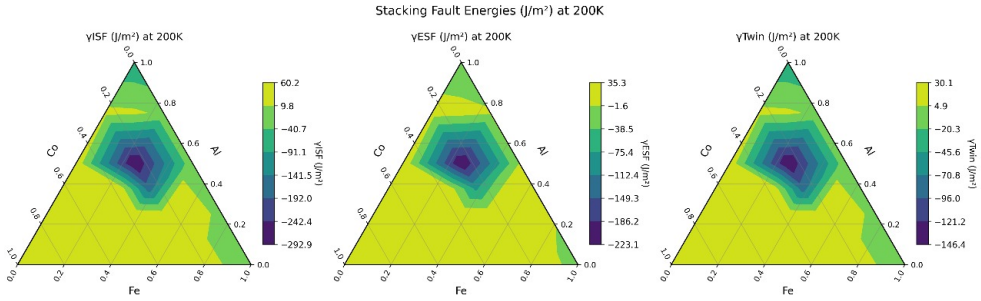


Figure 3: Variation of stacking fault energy (SFE) of Al–Co–Fe alloy at 200K

- The compositional trends displayed in the DMLF model are **smooth** and the f**ault energy evolution with temperature** has been represented correctly.
- The ANNNI model, including the **next-nearest-layer coupling predicts** a somewhat higher SFEs at lower temperatures and also generates oscillations of (γ_{ISF}), because of the interlayer interaction effects.
- At **650 K**, both models meet, which states the constant stabilization of the FCC phase.

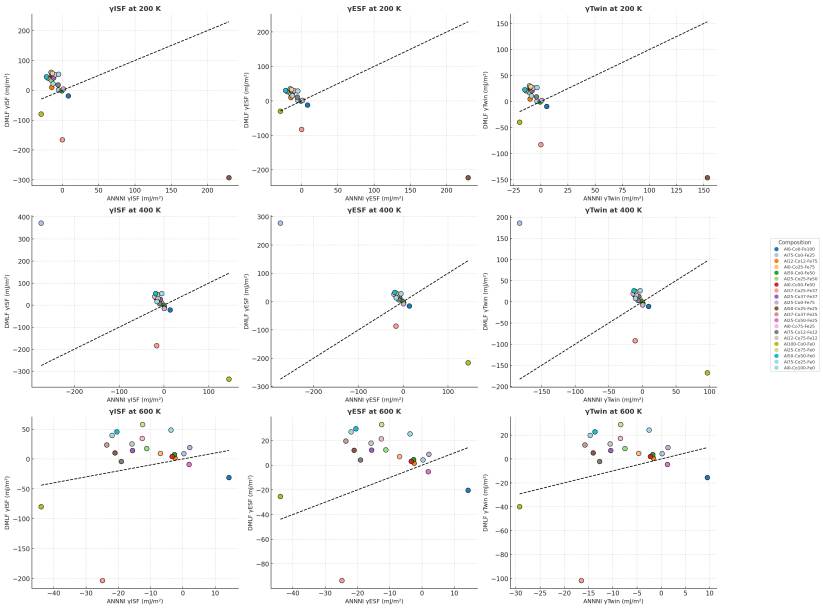


Figure 4: DMLF vs ANNNI at different compositions and temperatures

3.5 Phase Stability and Ternary Mapping

As we can see in Figure-5;

- Co- and Fe-rich alloys are inclined to stabilize in **HCP or DHCP** stacking at **low temperatures** and stabilize **FCC** at Al-rich regions at **high temperatures**, which has a macroscopic impact on the alloy.
- Upon increasing temperature, **phase boundaries move to the FCC region** which is in line with thermodynamic stabilization of high-symmetry structures.

4. Conclusions

- The **Al-Co-Fe** system combines excellent composition-temperature interdependence in stacking fault energies.

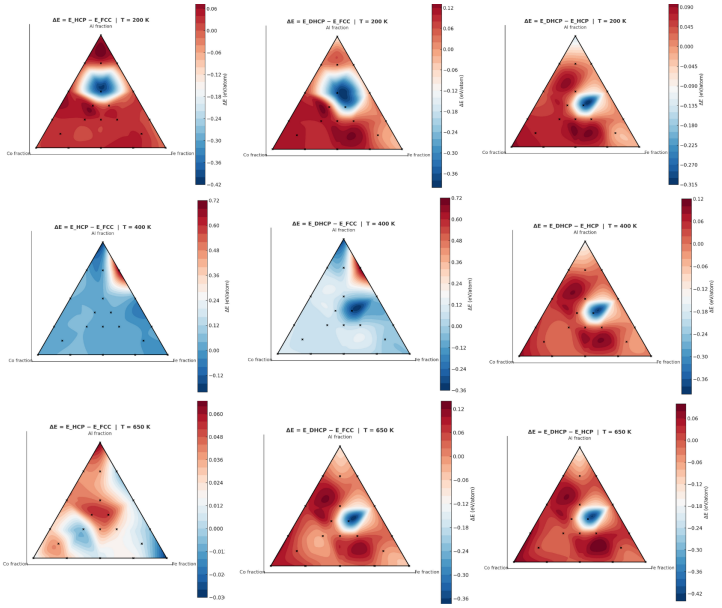


Figure 5: Phase Stability between different phases at different temperatures

- Al addition **increases** the lattice parameter and reduces cohesive energy which enhances planar faulting and FCC stabilization.
- The presence of Fe and Co **enhances bonding** and results in an **increase** in the SFEs and prefers the stacking of HCP/DHCP at lower temperatures.
- The rise of the temperature **elevates** the values of (γ_{ISF}), (γ_{ESF}), and (γ_{Twin}), which prove the **improved stability** of the ideal FCC lattice at higher temperatures.
- The **qualitative trends** of DMLF and ANNNI types are similar, with DMLF having more **continuous thermal transitions**, and ANNNI models **localized changes in stacking**.
- Phase maps verify a **transition to FCC dominance** as both temperature and Al concentration **increase**, as expected experimentally.

References

[1] M.A. Charpagne, K.V. Vamsi, Y.M. Eggeler, S.P. Murray, C. Frey, S.K. Kolli, and T.M. Pollock. Design of nickel–cobalt–ruthenium multi-principal element alloys. *Acta Materialia*, 194:224–235, 2020.

[2] S. Mahajan and G.Y. Chin. Formation of deformation twins in fcc crystals. *Scripta Materialia*, 46(9):653–659, 2002.

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[5] M. Hiroshi. mpltern documentation. https://mpltern.readthedocs.io/en/latest/index.html, 2024.